AI CLUSTERS

Automating AI Data Center Design with CLARA

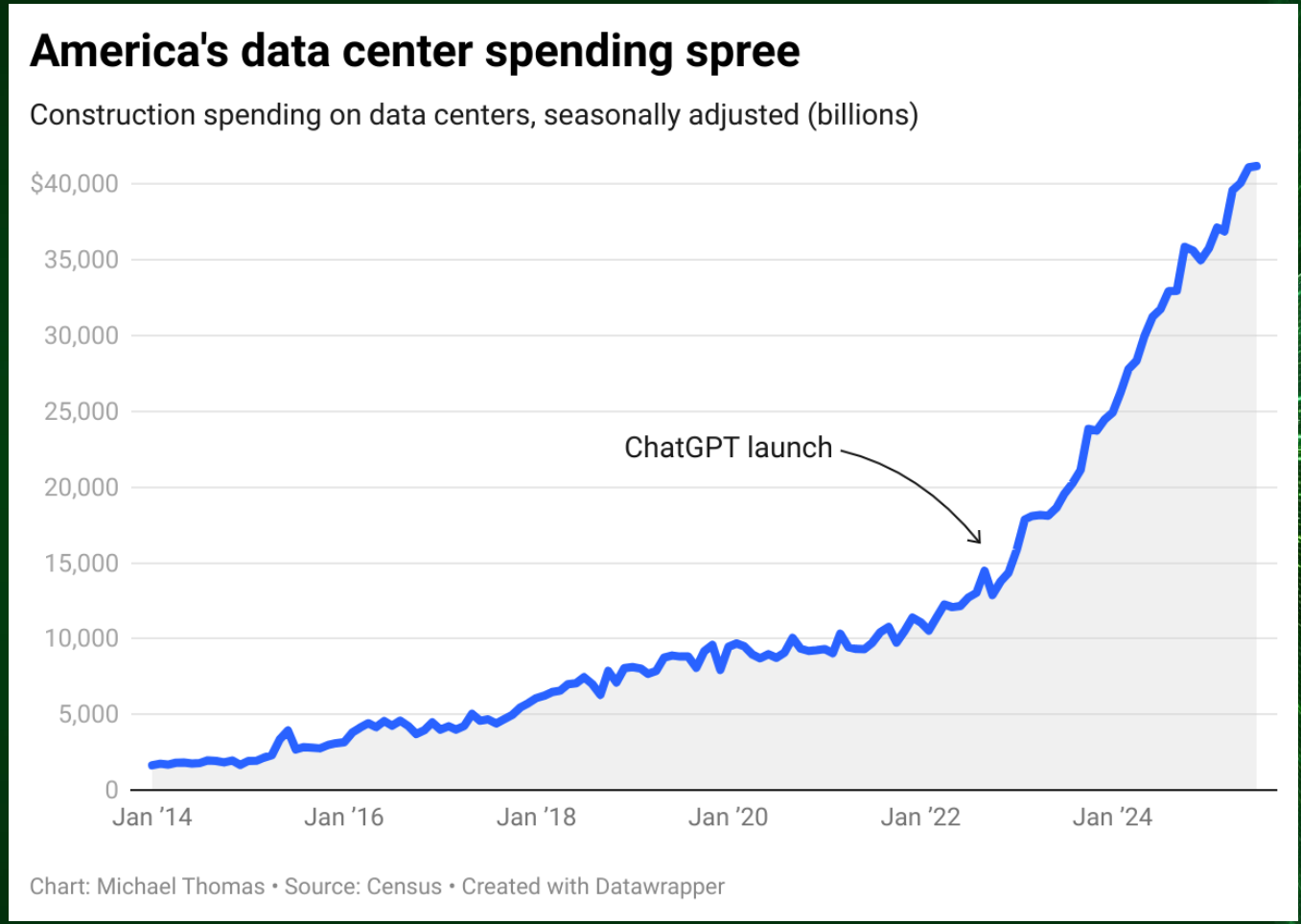
Rainy Yan, Reece Neff, Hengze Ye, Jerry Cheng

Infrastructure Solutions Group, Lenovo (CSP)

OCP BRONZE™ Member

Background

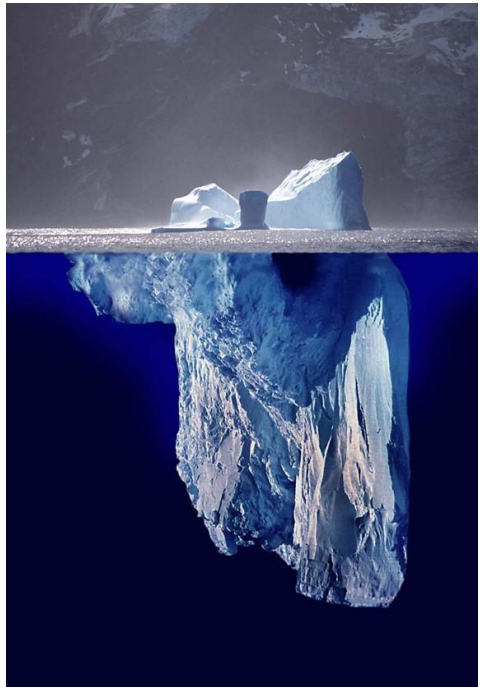
- Explosive growth of customized datacenter design requests
- Detailed cluster design requires a lot of manual effort to produce and reconciliate across different expert and vendors
- Static design and optimization methodologies cannot adapt to a fast and dynamic market



Main Problem: Lack of Holistic and Opensource Tool

- The ability to optimize cluster structure and cost is crucial for efficient ownership and operation
- Lack of specialized tools with holistic operation, optimization, and constraint checking for AI clusters
- Design intent is scattered across different documents, domains, vendors and deployment models

We need a unified, specialized, open ecosystem-based AI cluster design framework!



GPU and power cost are only 50% of the TCO – switch and cables are becoming the dominant cost factor in modern datacenters

Konstantin S. Solnushkin. 2014. Automated Design of Computer Clusters. Doctoral dissertation. Technische Universität München. Retrieved from tum.de

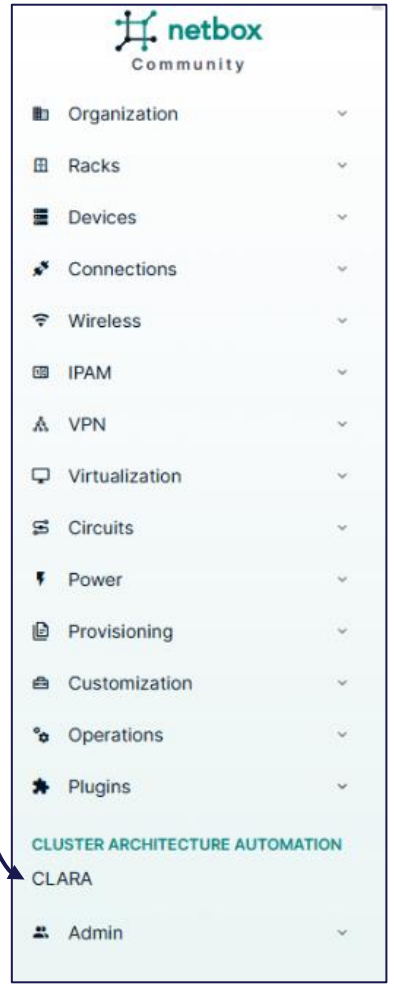
Image by Uwe Kils (iceberg) and Wiska Bodo (sky). Source: Wikimedia Commons.

Our Solution: CLARA

CLARA
CLuster ARchitecture Automation

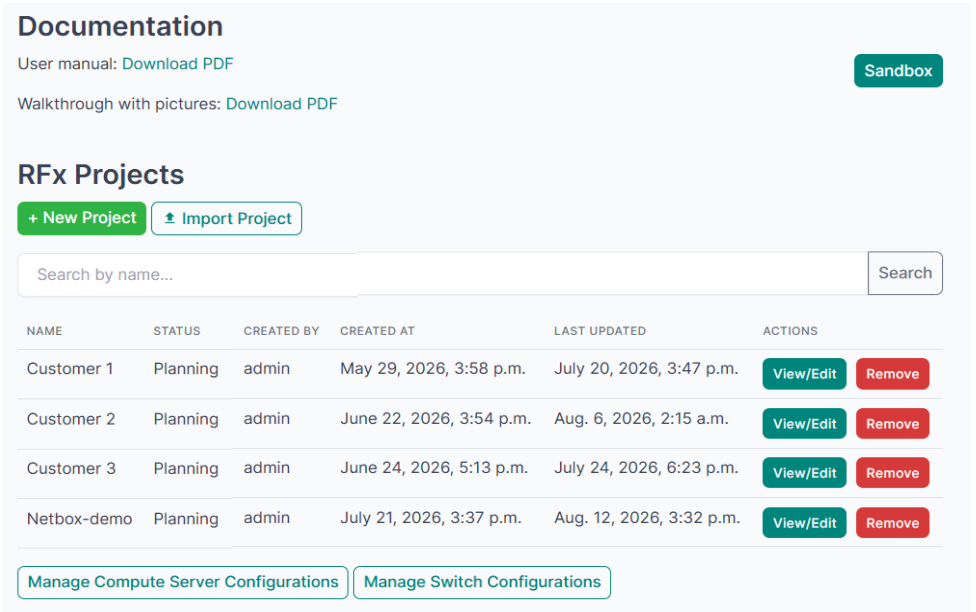
A plugin for Netbox, a widely-used open-source DCIM platform

- Preserves operational familiarity, eases adoption by the open-source community
- …while enhancing the automated OCP compliant or vendor preferred cluster design

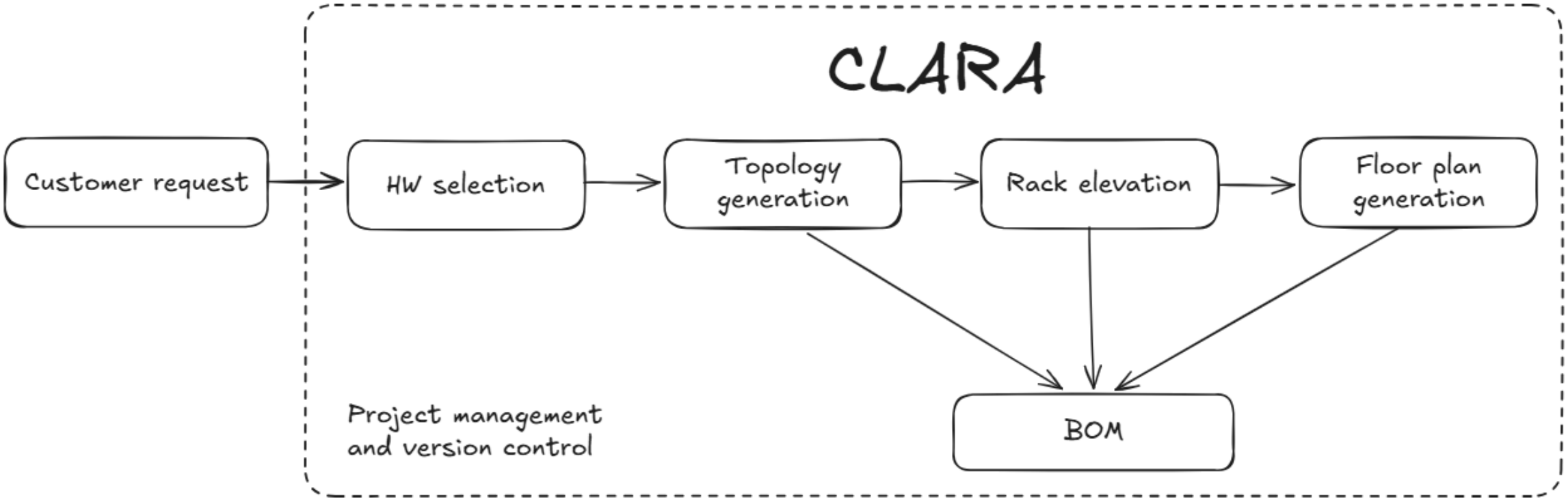


CLARA Objectives:

- Unified open-source ecosystem
- Automated constraint checking
- Modern AI cluster topology generation
- Automatic floorplan optimization



Unified Open-Source Tool



Design Workflow for AI Clusters

Hardware customization

- Reusable device configurations

Step-by-step workflow enforcement

- Modules are blocked unless all necessary information have been acquired

Automatic constraint checking

Target Device Quantities In Cluster

Compute Server: 0/2304

Backend Leaf: 0/320	Backend Spine: 0/1440	Backend Core: 0/1008
Frontend Leaf: 0/80	Frontend Spine: 0/40	Frontend Core: 0/32
CDL Server: 0/64	CDL Leaf: 0/66	CDL Spine: 0/8
HPS Server: 0/96	HPS Leaf: 0/66	HPS Spine: 0/16
OOB Leaf: 0/200	OOB Spine: 0/8	OOB Core: 0/2

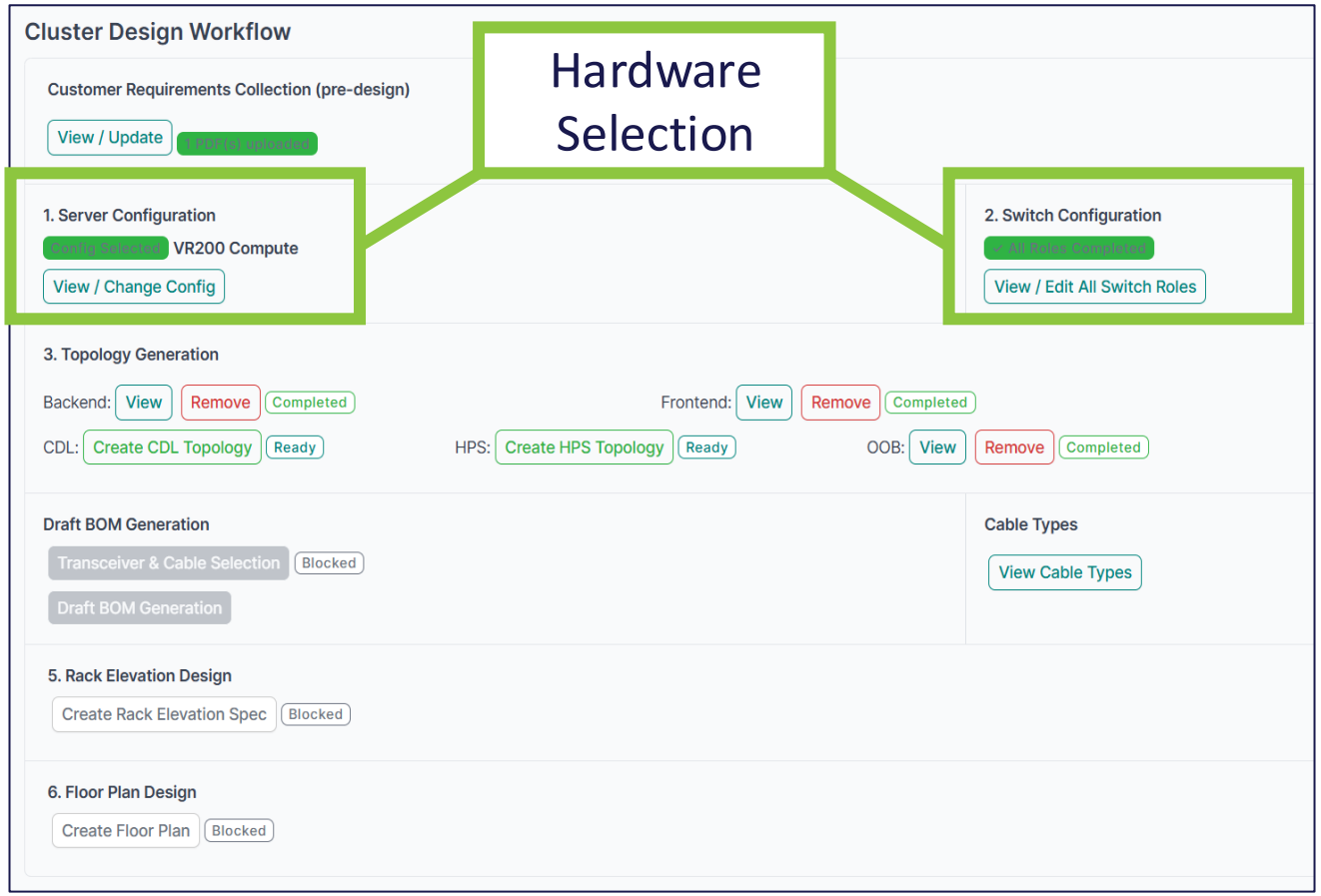
Current rack elevation info

Rack stats

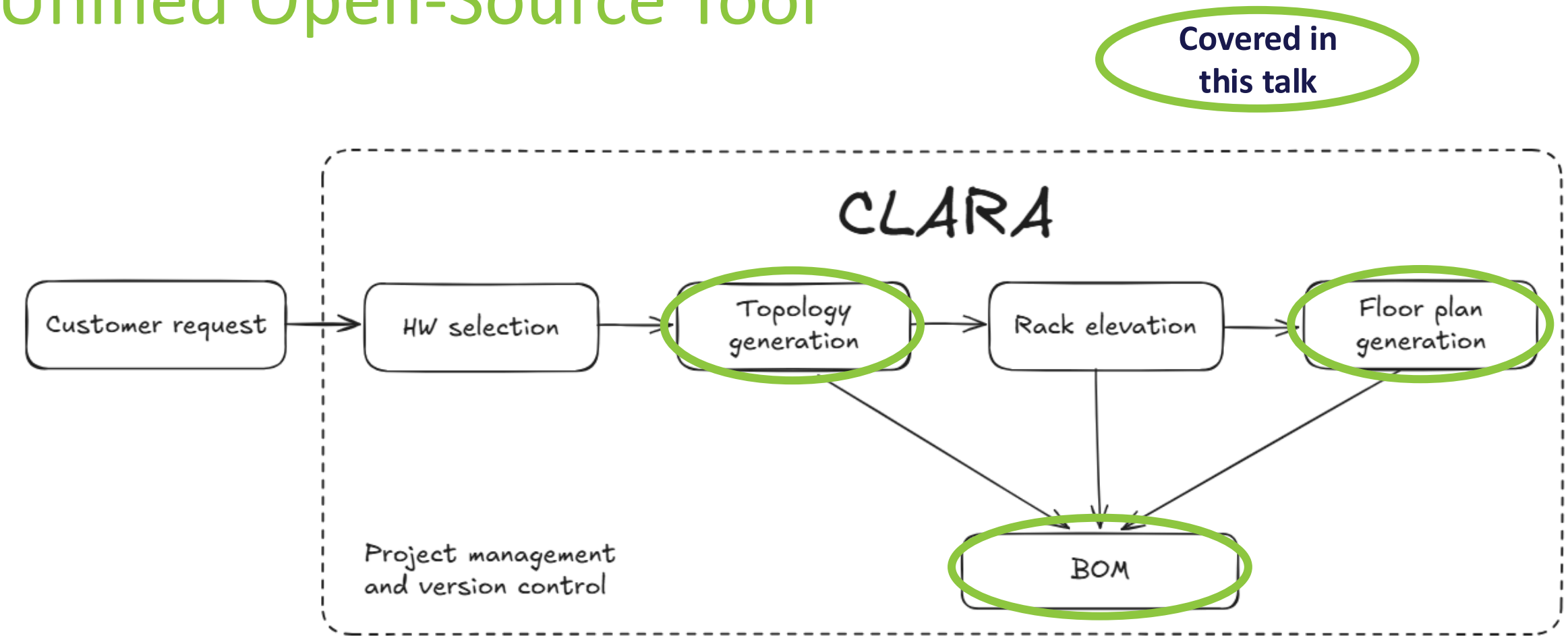
Space usage: 18/48 Power usage: 0/13500 Airflow usage: 0

Rack devices

GB300 Compute Tray (18) Compute Node: 18



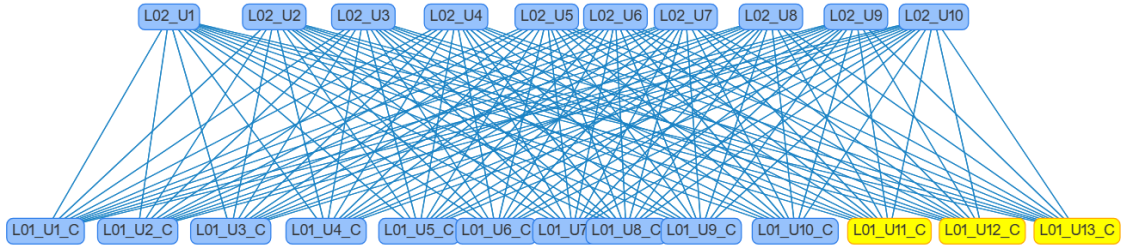
Unified Open-Source Tool



Problem 1: Physical Network Complexity

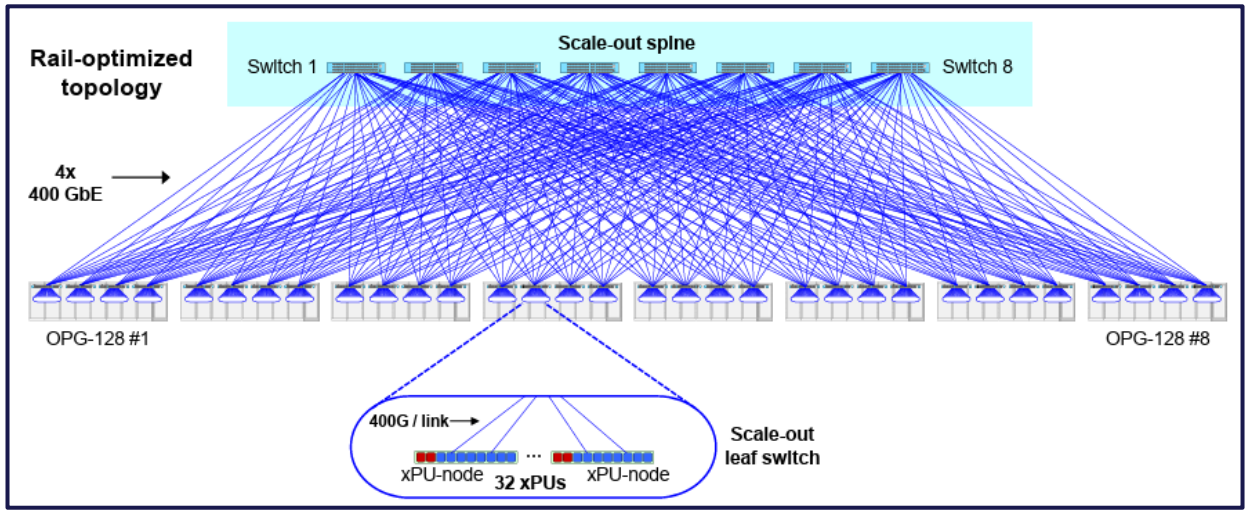
Existing topology tools rely on simple factory style, don't meet the demand of current **multi-rail, multi-plane** designs.

We need automatic topology generation for modern AI clusters!



CLUSTER CONFIGURATION TO USE	Switch	Fully Configured	Partially Configured		Total
			Switch Count	Port Count per Switch	
Level 1 Switches - Compute	40-port Switch	12	1	36	13
Level 2 Switches/Chassis	40-port Switch	0	10	26	10

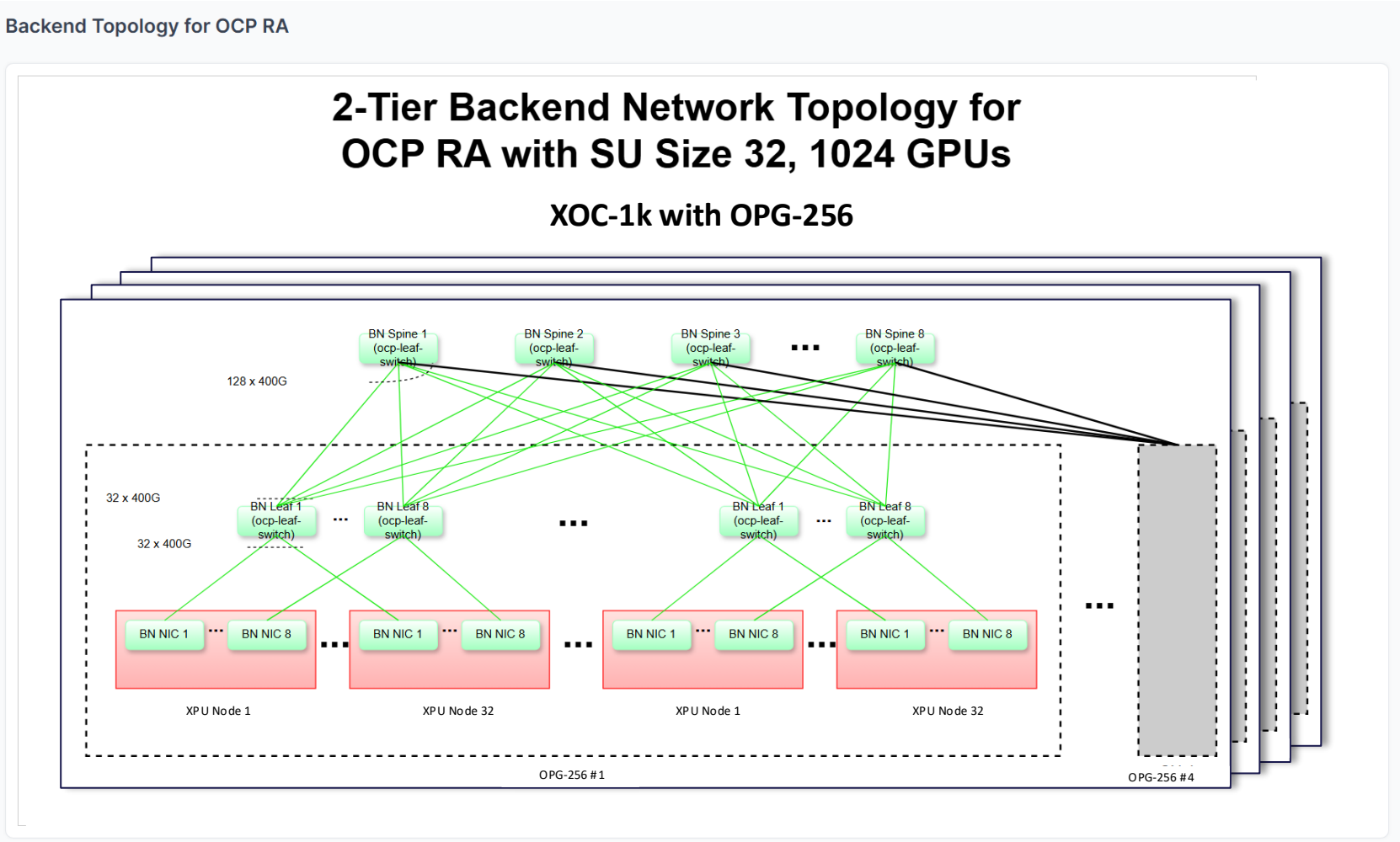
https://www.nvidia.com/en-us/networking/infiniband-configurator/



https://www.opencompute.org/documents/xoc-n-system-architecture-final-14-january-2026-pdf

CLARA's Network Topology Algorithms

- Rail optimized, multi-plane
- Node- and switch-aware
- Searches only feasible topology configurations

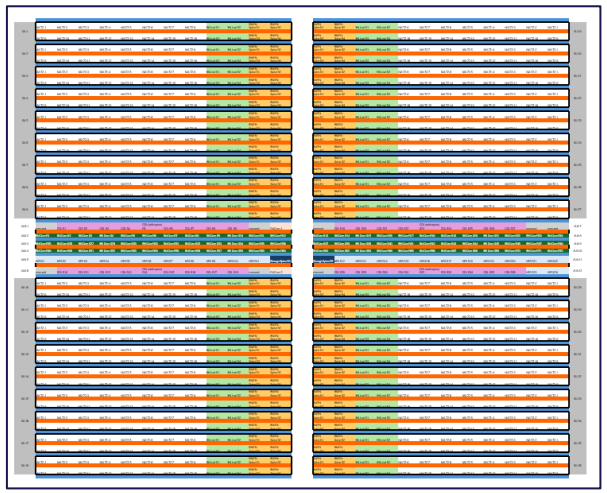


Problem 2: Floorplan Complexity

- Geometry (rooms, floors, columns)
- Infrastructure reach (networking, CDU pipes, power)
- Operational preferences
- *Different definitions of "Optimal"*

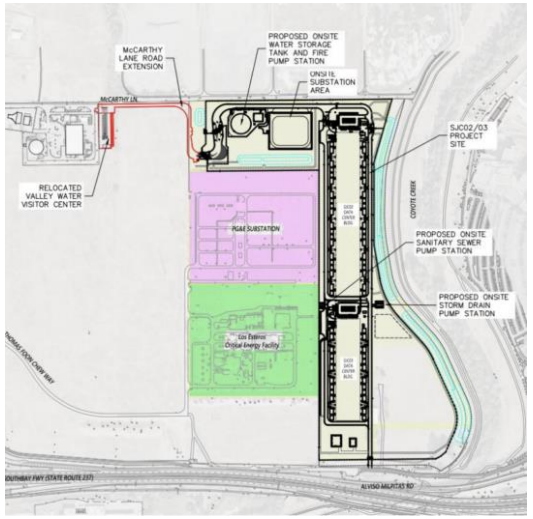
We need an approach for automatic floor planning and cable length estimation!

Reference Architecture



Single Hall, Perfect Square

Real-World Examples



Multi-Building Design

https://sfyimby.com/2025/04/microsoft-data-center-approved-by-san-jose-planning-commission.html



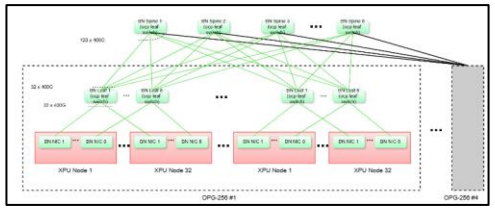
Multi-Floor Design

https://www.serverfarmllc.com/build-to-suit-data-centers/

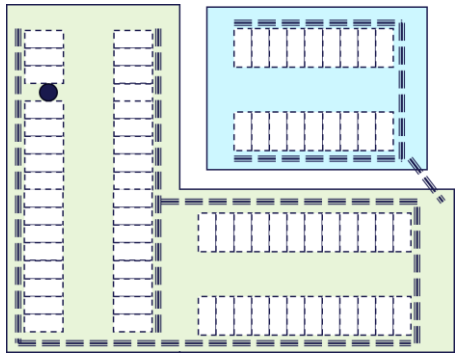
CLARA's Automatic Floorplan Generation

Input

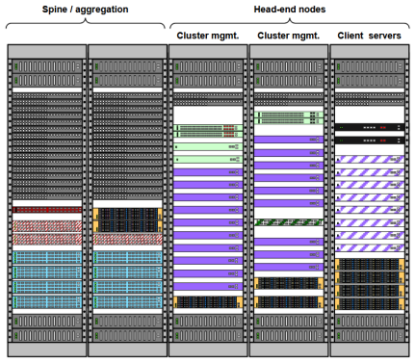
Network Topology



Floorplan Template

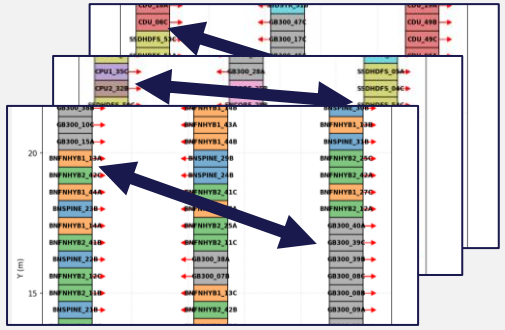


Rack Elevations



Optimization

Simulated Annealing



Optimization Time
< 10 minutes

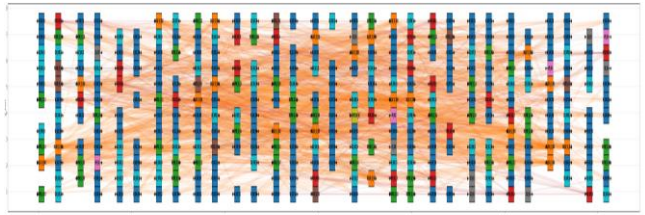
Preliminary Cost Function

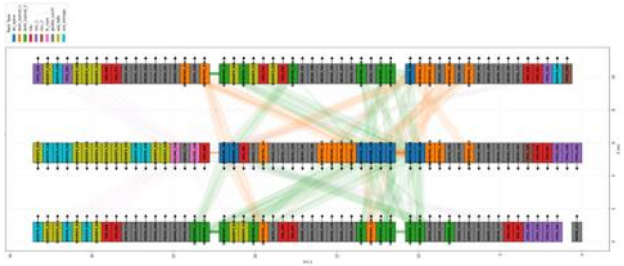
$$\min C_{total} = \alpha C_{net} + \beta C_{cdu}$$

$\boldsymbol{C_{net}} = network\ length$
$\boldsymbol{C_{cdu}} = CDU\ length$
$\alpha, \beta = weights$

Output

Optimized Floorplans





Automatic BOM Generation

- Auto calculates inter- and intra-rack cable lengths from rack elevations and floorplan
- Exports to .xlsx/.csv formats
- Automatic cable/transceiver compatibility filtering

CONNECTION ROLE	A-SIDE	B-SIDE	USE TRANSCEIVER *	A-SIDE TRANSCEIVER	B-SIDE TRANSCEIVER	CABLE TYPE	ACTION
frontend nic leaf	400gbase-x-qsfp112 1 x 400	800gbase-x-osfp 2 x 400	Yes	Generic Compatible 400GBASE-SR4 QSFP112 PAM4 850nm 50m DOM MPO-12/APC MMF Optical Transceiver Module	Twin-Port Transceiver, 800Gbps, OSFP, Multi Mode Twin-Transceiver, 2xMPO12 APC, 850nm MMF, Up to 50m, Finned	30M MPO12 APC Female to MPO12 APC Female, MMF, OM4, Type B Fiber Cable	Select

Bill of Materials (Draft)

#	CATEGORY	MANUFACTURER	MODEL	PART NUMBER	QUANTITY	COST
1	**Compute Server**	Vendor 1	GPU Node	123	2520	0
2	**CDL Server**	Vendor 2	CPU Node		38	0
3	**HPS Server**	Vendor 2	CPU Node		256	0
4	**Backend Leaf Switch**	Vendor 3	64x800GbE Switch SONiC, Air Cooled		80	0
5	**Backend Spine Switch**	Vendor 3	64x800GbE Switch, SONiC, Air Cooled		42	0
6	**Frontend Leaf Switch**	Vendor 3	64x800GbE Switch, SONiC, Air Cooled		80	0
⋮						
21	**Frontend Nic->Leaf Transceiver**	Vendor 4	Generic Compatible 400GBASE-SR4 QSFP112 PAM4 850nm 50m DOM MPO-12/APC MMF Optical Transceiver Module		5120	0
22	**Frontend Leaf->Nic Transceiver**	Vendor 4	Twin-Port Transceiver, 800Gbps, OSFP, Multi Mode Twin-Transceiver, 2xMPO12 APC, 850nm MMF, Up to 50m, Finned		5120	0
23	**Frontend Nic-Leaf Cable**	Vendor 5	30M MPO12 APC Female to MPO12 APC Female, MMF, OM4, Type B Fiber Cable		5120	0

Call to Action

- Estimated open-source availability: **February 2027**
  - Try it out & contribute whenever it becomes available!

- Add devices to open-source NetBox Device Type Library and NetBox Data Exchange
  - https://github.com/netbox-community/devicetype-library
  - https://netboxlabs.com/ndx/

- Participate in Open Cluster Designs for AI Workgroup meetings
  - https://www.opencompute.org/community/cluster-designs-for-ai

- Contact the team (clara@lenovo.com) for collaboration opportunities and early access


Thank You!
Smarter technology for all
Lenovo
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San Jose, California